

Grouping of Images

COMPLETE LEARNING GUIDE

What You'll Learn

Grouping-of-images (grouping identical figures) questions give a jumbled set of figures — usually the same handful of distinct base shapes repeated at different rotations and sizes — and ask you to sort them into groups where every figure in a group is truly the same shape, just rotated or resized, while figures that only look superficially similar but are structurally different (especially mirror images) must be kept separate. This sub-module builds a rotation-invariant comparison method: tracking turn sequences to detect true rotations versus mirror images, using a fixed reference feature to track orientation consistently, and handling figures with internal details like arrows or shading whose relative position must also match.

1. The Core Task: Sorting by True Identity, Not Surface Appearance

Grouping-of-images questions give a jumbled set of figures — usually the same handful of distinct base shapes repeated at different rotations and sizes — and ask you to sort them into groups where every figure in a group is actually the SAME shape (just rotated or resized), while figures that only look superficially similar but are structurally different must be kept separate.

- Nine arrow-like figures are given. Visually they all look like 'a bent arrow,' but structurally there are only 3 distinct bend angles among them (some bend at 90° , some at 120° , some at 60°), each appearing 3 times at different rotations. The correct grouping is 3 groups of 3, sorted by bend angle, not by which way each arrow currently points.

2. Rotation-Invariant Comparison

To check if two figures are the same shape rotated differently, compare properties that don't change under rotation — the sequence of angles going around the shape, the relative lengths of its sides in order, and how many 'arms' or protrusions it has — rather than comparing their orientation on the page, which is expected to differ within a valid group.

- Two zigzag figures each have 3 straight segments. Figure X's segments turn left, then right, then left (reading from one end to the other). Figure Y, rotated 180° from X, still turns left, then right, then left when read consistently from the same relative end — confirming they're the same shape, just rotated.

3. Detecting Mirror Images (Which Are NOT the Same Figure)

A mirror-image version of a shape is NOT considered identical for grouping purposes even though it has the same angles and side lengths — check this by tracing the shape's sequence of turns (left-turn vs. right-turn at each bend) in a consistent direction; a true rotation preserves the exact left/right turn sequence, while a mirror image reverses every left turn to a right turn and vice versa.

- Figure X's zigzag turns left, then right, then left (reading clockwise from one end). A candidate Figure Z has the same 3 segment lengths and angles, but reading it the same way gives right, then left, then right — the exact reverse sequence. Figure Z is a MIRROR image of X, not a rotation, and must NOT be grouped with it.

4. Using a Reference Feature to Track Orientation

Pick one distinguishing feature on the base shape (like its longest side, or a small notch/arrow marking) and use it as a fixed reference point when comparing figures — mentally rotate each candidate figure until its reference feature aligns with the reference feature of a known group member, then check whether the rest of the shape lines up too.

- A shape has one side noticeably longer than the other three. Using that longest side as the reference, rotate a candidate figure mentally until its longest side points in the same direction as the reference figure's longest side — if every other feature then lines up exactly, they're the same shape; if not, they're different.

5. Grouping Figures With Internal Details

When figures include internal details (an arrow inside a shape, a shaded portion, a small dot), those details must maintain the SAME relative position to the outer shape's reference feature across the whole group — an outer shape that matches perfectly but has its internal arrow pointing toward a different relative side is a different figure for grouping purposes, not the same one misdrawn.

- Two identical pentagons each contain an internal arrow. In the reference figure, the arrow points from the center toward the longest side. In a candidate figure with the same pentagon outline, the arrow instead points toward the shortest side. Despite the identical outer pentagon, this candidate belongs to a DIFFERENT group because its internal detail's relative position doesn't match.

6. Worked Full Example: Sorting 9 Figures Into 3 Groups of 3

Work through a full sort methodically: pick any ungrouped figure as a reference, use its rotation-invariant properties (turn sequence, relative side lengths, reference feature) to find its 2 matches among the remaining figures, remove all 3 from the pool, and repeat with a fresh reference figure from what's left.

- 9 figures are given: 3 are '4-pointed star' shapes at various rotations, 3 are 'plus-sign' shapes at various rotations, and 3 are 'pinwheel' shapes (which look star-like at a glance but have curved, not straight, blade edges). Picking one star as reference and checking turn-sequence and edge-type (straight vs curved) correctly separates the 3 true stars from the 3 pinwheels, which a quick glance might otherwise lump together as the same shape.

7. Distinguishing Near-Identical Decoys

The most common trap figure in a grouping set differs from a true group member in exactly one small way — one side very slightly shorter, one angle very slightly wider, or one extra small element — designed to be caught only by careful comparison of the specific measurable features, not by a quick visual scan.

- Two 'house-shaped' pentagons (a square with a triangular roof) look identical at a glance. Careful comparison shows one has a roof triangle that's a true equilateral shape (all angles 60°), while the decoy's roof is slightly flatter (a wider, shorter triangle) — a difference easy to miss without directly comparing the roof's proportions, not just its general silhouette.

Common Pitfalls

- Grouping figures by rough visual similarity instead of verifying rotation-invariant properties like turn sequence and relative side lengths.
- Accidentally grouping a mirror image with its true rotational match, since mirror images share angles and side lengths but reverse the turn sequence.
- Ignoring internal details (arrows, shading, dots) and matching only on the outer silhouette.
- Missing that an internal detail's relative position (not just its presence) must match across a group.
- Being fooled by a decoy figure that differs by only one slightly altered angle or side length.
- Not using a fixed reference feature to track orientation, leading to inconsistent 'by eye' comparisons across many rotated figures.

Key Takeaways

- Compare figures using rotation-invariant properties (turn sequence, relative side lengths) rather than their orientation on the page.
- A mirror image reverses the left/right turn sequence and is NOT the same figure for grouping purposes, even with identical angles and lengths.
- Internal details (arrows, shading, dots) must match in both presence AND relative position to the shape's reference feature.
- Use one clear reference feature (longest side, a notch) to mentally align and compare rotated candidates consistently.
- Sort methodically: fully resolve one group at a time using a fresh reference figure from the remaining pool.

- Watch for decoy figures differing by only one small, easy-to-miss measurement rather than an obviously different shape.

Learn More

IndiaBix — Non-Verbal Reasoning: Grouping Identical Figures

<https://www.indiabix.com/non-verbal-reasoning/grouping-identical-figures/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>